🔑 **Answer Key — Teacher Copy · Lesson 5-2**

MSTAR PRACTICE WORKSHEET

Lesson 5-2: Determine the Area of Triangles

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

What is the area of a triangle with base 10 cm and height 6 cm?

A. 16 sq cm

B. 30 sq cm

C. 30 cm

D. 60 sq cm

A = ½ × b × h = ½ × 10 × 6 = 30 square centimeters.

Question 2 SELECTED RESPONSE — CORRECT: B

A triangle has an area of 24 sq ft and a base of 8 ft. What is the height?

A. 3 ft

B. 6 ft

C. 12 ft

D. 16 ft

A = ½ × b × h → 24 = ½ × 8 × h → 24 = 4h → h = 6 ft.

Question 3 SELECTED RESPONSE — CORRECT: A

Why do we divide by 2 when finding the area of a triangle?

- (A) A triangle is exactly half of a rectangle with the same base and height
- (B) Triangles have 2 equal sides
- (C) The base is always twice the height
- (D) We always divide area formulas by 2

A triangle with base b and height h fits inside a rectangle with the same base and height. The triangle covers exactly half the rectangle, so A = ½ × b × h.

Question 4 SELECTED RESPONSE — CORRECT: B

What is the area of a triangle with base 14 ft and height 6 ft?

- (A) 20 sq ft
- (B) 42 sq ft
- (C) 42 ft
- (D) 84 sq ft

A = ½ × 14 × 6 = ½ × 84 = 42 sq ft.

Question 5 SELECTED RESPONSE — CORRECT: B

A triangular window in the firm's lobby has a base of 10 ft and a height of 6 ft. What is its area?

- (A) 16 sq ft
- (B) 30 sq ft
- (C) 32 sq ft
- (D) 60 sq ft

A = ½ × base × height = ½ × 10 × 6 = ½ × 60 = 30 sq ft. The triangle is half of the 10 × 6 = 60 rectangle it fits inside.

Question 6 SELECTED RESPONSE — CORRECT: B

A triangular roof panel on the blueprint has a base of 7 ft and a height of 4 ft. What is its area?

A. 11 sq ft

B. 14 sq ft

C. 22 sq ft

D. 28 sq ft

A = ½ × base × height = ½ × 7 × 4 = ½ × 28 = 14 sq ft. Multiply base × height first (28), then take half (14).

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A triangular garden bed has a base of 14 meters and a perpendicular height of 6 meters. What is its area?

A. 10 square meters

B. 20 square meters

C. 42 square meters

D. 84 square meters

Area = ½ × base × height = ½ × 14 × 6 = 42 square meters.

- **A:** That halves the sum of 14 and 6. Multiply first: half of 14 times 6.
- **B:** That adds 14 and 6. Multiply the two measurements, then halve.
- **D:** That is base times height with no halving, which gives the PARALLELOGRAM area. A triangle is half of it.

Part B — correct answer: A

Which statement explains why the area formula for a triangle includes the factor ½?

A. A triangle is exactly half of a parallelogram with the same base and height.

B. Triangles always have three sides, so you divide by three and round.

C. The height of a triangle is always half of its base.

D. Half of any shape's area is its perimeter.

Two identical triangles fit together to form a parallelogram with the same base and height, so one triangle is half of b · h.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: The Missing One-Half

> Dev found the area of a triangle with a base of 10 cm and a height of 7 cm. He wrote: 'Area = 10 × 7 = 70 square centimeters.'

Explain Dev's misconception and give the correct area.

> **Model answer:** Area = ½ × 10 × 7 = 35 square centimeters. Dev used b · h, which is the area of a parallelogram; a triangle is half of that parallelogram.

2 points	Student identifies that Dev used the parallelogram formula and omitted the factor of ½, and gives 35 square centimeters.
1 point	Student gives 35 square centimeters but does not explain why the ½ is needed.
0 points	Student agrees with Dev or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A rectangle is 12 ft by 8 ft. A diagonal line cuts it into two triangles. What is the area of each triangle? Explain how the triangle area formula relates to the rectangle area formula.

> **Model answer:** The rectangle's area is 12 × 8 = 96 sq ft, and one diagonal cuts it into two identical triangles, so each triangle is 96 ÷ 2 = 48 sq ft. The formula says the same thing: each triangle keeps the rectangle's base of 12 and height of 8, and A = ½ × 12 × 8 = 48 sq ft. The ½ in the triangle formula IS that diagonal cut.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.